



SMBA Draft RFP Amendment 18 Industry Day

January 2026



- Purpose: Provide an overview of SMBA PRD changes
 - Note: The final RFP is the next major drop. While there may be additional Industry Days and some smaller updates, no additional full dRFP update is planned.
- Agenda
 - Ground Rules
 - SMBA RFP Updates to PRD
 - Timeline
 - Questions

- The NEON Series-1 team will continue to conduct one on ones with Industry Partners as requested until Black Out
 - Earlier feedback and discussions are most helpful
- Salient information will be provided to all Industry Partners
 - Questions will be posted anonymously
- The final chart deck from today is posted on SAM.gov
- The Final Request for Proposal and its terms and conditions will supersede any statements in this document

SMBA PRD NS1-SMBA-REQ-0001 Rev D

Change Overview

- Updates to the SMBA PRD are in response to:
 - Industry partner responses to:
 - Draft RFP posted 4/18/2025 (NS1-SMBA-REQ-0001 Rev. B)
 - Draft RFP Amendment 13 posted 9/5/2025 (NS1-SMBA-REQ-0001 Rev. C)
 - Internal NASA technical discussions
 - NASA program and project changes
- Changes from SMBA PRD Rev C to Rev D affect all sections of the PRD.
 - Major performance, function, and test requirement revisions are on the next slides
 - Other changes clarify the PRD requirement set and do not significantly change the PRD intent (e.g., revising requirement language, adding context, restructuring sections, and moving requirements to SOW/CDRL)
 - Requirements are renumbered in sequential order
 - No new TBXs affect instrument design, interface with spacecraft, or testing
 - Revised TBX list and added closure event
 - Editorial changes throughout

SMBA PRD Performance Requirements Change Summary

PRD Sect.	PRD Rev D Req ID	PRD Rev C Req ID	Description of change
4.1	Table 4.1-2	Table 4.1-2	Changes to Hyperspectral Resolution column - Added one decimal place to each value (e.g., from 10 MHz to 10.0 MHz) - Relaxed the V-band narrowest hyperspectral resolution to only be around the peaks of the oxygen absorption spectrum - V band Hyperspectral Resolution ≤ 1.0 MHz for spectrum ≤ 6 MHz from any absorption peak and ≤ 10.0 MHz elsewhere.
4.1	SMBA-7	SMBA-6	Refined the aspects of the hyperspectral and RFI detection capability that are required to be reprogrammable On-orbit instrument capability to reprogram the number of channels, number of passbands, center frequency of each passband, bandwidth of each passband, and frequency resolution for all hyperspectral and RFI detection bands
4.1	Table 4.1-3	Table 4.1-3	Changes Heritage Channel requirements - Fixed typos in channel center frequencies to align with Advanced Technology Microwave Sounder (ATMS) heritage - Center frequency change from 53.71080 GHz to 53.481 GHz (ATMS Ch 6) - Center frequency change from 54.48080 GHz to 53.711 GHz (ATMS Ch 6) - Center frequency change from 57.073350 GHz to 57.073344 GHz (ATMS Ch 11) - Center frequency change from 57.507340 GHz to 57.507344 GHz (ATMS Ch 11) - No longer requiring heritage channels in unprotected spectrum - Removed 51.76 GHz (ATMS Ch 4) and modified lowest F-Band channel center frequency from 114.5 GHz to 114.75 GHz - Added new heritage channels to cover the missing Micro-wave Sounder (MWS) channels - Center frequency 53.246$\pm$0.08 GHz (MWS Ch 5), 53.948$\pm$0.081 GHz (MWS Ch 7), and 89 GHz (MWS Ch17) - Added new channel bandwidth tolerance of +/- 5% - Added new column to Table 4.1-3 for beam polarization that aligns with heritage instruments

SMBA PRD Performance Requirements Change Summary

PRD Sect.	PRD Rev D Req ID	PRD Rev C Req ID	Description of change
4.8.1	SMBA-34 SMBA-35 SMBA-36 SMBA-37	N/A	Added new beam polarization requirements applicable to all SMBA channels - A minimum of one linear polarization is required - Polarization twist angle < 0.1 degree - Cross-polarization leakage $< -20\text{dB}$ for K, Ka, W band, and $< -15\text{dB}$ for all other frequency bands - See Q&A for more detail
4.2	SMBA-13 SMBA-14	N/A SMBA-14	Requiring RFI detection packet instead of flag - SMBA RFI detection packet provides the information to compute metrics on RFI detection algorithm performance. Ground processing will produce science data quality flags and metrics based on the RFI detection packet. - RFI detection packet identifies the algorithm that detected the RFI, threshold exceedance, frequency range corrupted, channel(s) affected, and time stamp.
4.3	Table 4.1-4 All reqs revised	Table 4.1-4 All reqs revised	Revised Section 4.3 Calibration Accuracy and is now retitled System Calibration Uncertainty - Aligned Calibration Uncertainty definition, contributors, and accounting with NIST, BIPM, and JPSS. - Rev. C defined calibration accuracy as a function of magnitude and uncertainty - Rev. D defines calibration uncertainty as a function of accuracy and precision, no longer uses static and dynamic terminology, requirements are on accuracy and uncertainty - Generally, uncertainty requirements apply on-orbit, after bias corrections - Revised notional contribution error sources in Figure 4.3-1. (see next slide) - Rev D Table 4.1-4 now requires the total calibration accuracy (see next slides) - Deleted Rev C SMBA-26 and SMBA-27 near-field and far-field antenna characteristics requirements - Orbital variation of accuracy for any single orbit < 0.2 K - Lifetime stability of orbital-mean accuracy < 0.2 K - Maximum calibration accuracy difference across the scan < 0.3 K. Note: Scan angle dependent bias characterizes the consistency of calibration accuracy within a scan. - Magnitude of antenna temperature accuracy of the warm calibration reference < 0.20 K - Magnitude of antenna temperature accuracy of the cold space calibration reference < 0.5 K for K, Ka, W bands, < 0.2 K for all other bands - See Q&A for more detail.

SMBA PRD Performance Requirements

Rev D Figure 4.3-1

Revised notional contribution error sources in Figure 4.3-1.

- Reminder Figure 4.3-1 is notional and not required
- Near-field contamination has been added to the warm calibration reference and cross-polarization leakage has been added to the radiometer calibration transfer function.
- To avoid double-counting the uncertainty due to NEDT, the Figure has removed "System Noise" from Radiometric Transfer Function and the total calibration precision excludes precision uncertainty from NEDT.

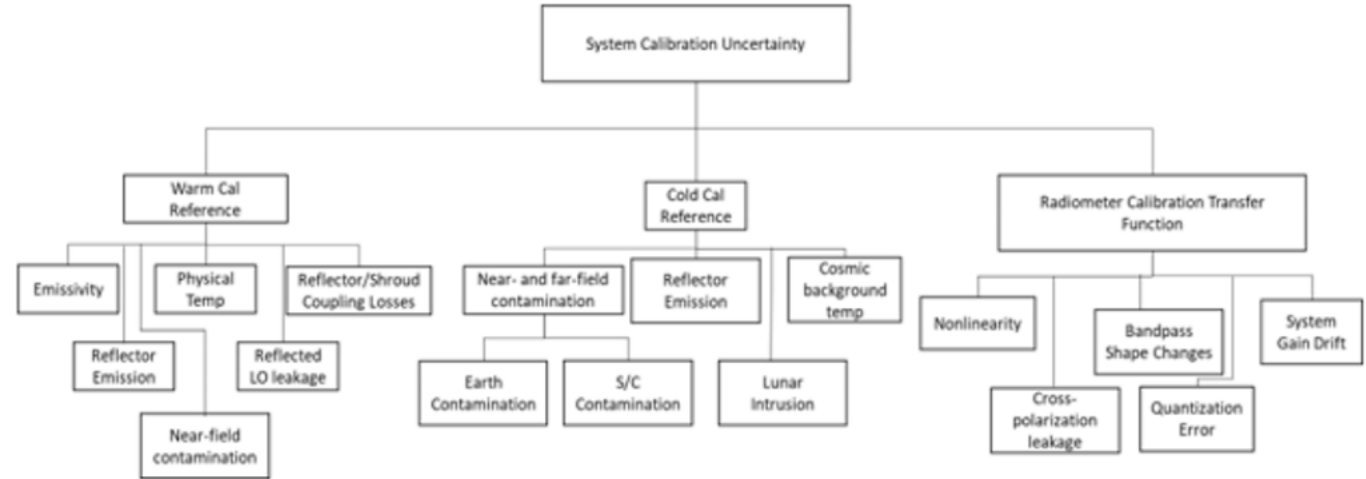


Figure 4.3-1. Notional Calibration Uncertainty Contributions

Notes:

- [1] A calibration reference is defined as an element that provides a signal that corresponds to a known brightness temperature. It can be a physical thing (like a blackbody; ex: on-board target or cold space). A noise diode is also an example of a reference.
- [2] Traditional radiometers have used two-point calibration methodology: cold space for the cold reference temperature and an on-board blackbody for the warm reference temperature. The instrument design is not constrained to this methodology or these calibration reference types. Other types of calibration references exist such as noise diodes and cold Field Effect Transistors (FET).
- [3] Proposed references that are also standards (as defined by the National Institute of Standards and Technology (NIST)) must include a description of the traceability plan that enables them to be a standard, along with an estimate of the (absolute) accuracy. SMBA follows the NIST definitions of "standard," "reference," "intercalibration," and "intercomparison."

Values for contributions in Figure 4.3-1 are characterized on the ground, on-orbit, and/or by analysis.

SMBA PRD Performance Requirements Change Summary

PRD Sect.	PRD Rev D Req ID	PRD Rev C Req ID	Description of change
4.4	Table 4.1-4 SMBA-25 N/A	Table 4.1-4 SMBA-28 SMBA-29	Redefined Radiometric Sensitivity (NEDT) as having two contributions: thermal (white) noise and 1/f noise (pink or flicker noise) - Thermal (white) noise contribution is scalable for bandwidth and integration time. - Rev D Table 4.1-4 now requires the thermal noise contribution only (see below) - Deleted Rev C SMBA-29 - See Q&A for more detail

Table 4.1-4. NEON Program Microwave Sounder Channel Performance Requirements

Frequency Bands	Center Frequency Stability (MHz)	Total Calibration Accuracy (absolute value of E_{acc}) (K)	Thermal Noise Contribution to NEDT ^{[1],[2]} (K)
K	5	1.0	15.8
Ka	5	1.0	15.1
V	0.3	0.5	17.7
W	18	1.0	36.0
F	1	0.5	29.4
D	22	0.5	57.9
G	9	0.5	63.6
J	22	0.5	64.8

Notes:

[1] Thermal Noise Contribution to NEDT values:

(a) Assume a 300K scene temperature observation.

(b) Apply to an integration time of 1 msec and a bandwidth of 1 MHz; Integration time is defined as the time interval over which the radiometric signal is collected for a given channel.

(c) Assume no radio frequency interference in measurement.

(d) Apply at the instantaneous field of view (IFOV). Instrument provider to show NEDT computed values from the IFOV and NEDT computed values after any resampling or aggregation.

[2] To scale the Thermal Noise Contributions to NEDT values for the actual integration time and bandwidth, divide the value by $\sqrt{\text{actual integration time in msec} \times \text{actual bandwidth in MHz}}$.

SMBA PRD Performance Requirements Change Summary

PRD Sect.	PRD Rev D Req ID	PRD Rev C Req ID	Description of change
4.7	SMBA-30	SMBA-33	Deleted striping requirement and replaced with 1/f noise ratio requirement 1/f noise ratio required to be ≤ 0.10 - See Q&A for more detail
4.8.4	SMBA-42	N/A	Added new passband gain ripple requirement - Peak-to-peak gain variation within each passband < 1.0 dB for K, Ka, and V; < 1.5 dB for W, F, D, G, and J bands - Applies over 80% of the passband bandwidth centered on the center frequency of the passband
4.8.5	N/A	SMBA-43	Deleted vertical reporting interval requirement from PRD Requirement for vertical reporting interval is maintained at project level requirements
4.9	SMBA-52 SMBA-53	SMBA-50 SMBA-51	Resolved the TBDs for SRF requirements - Magnitude (gain) knowledge of an SRF within 0.1 dB - The number of data points in SRF measurements ≥ 300 data points for each passband, including data points on the skirts and adjacent stopbands.

Functional and Test Requirements Change Summary

PRD Sect .	PRD Rev D Req ID	PRD Rev C Req ID	Description of change
5.3 5.5.1	SMBA-109 SMBA-110 SMBA-118	SMBA-109 SMBA-110 SMBA-117	Revised storage and maintenance periods - Maximum ground storage period extended to 13 years - Minimum periodic interval of 12-month period for maintenance during assembly, integration & test, and ground storage
5.7.2	SMBA-149	N/A	Added new instrument fundamental resonant frequency requirement (> 100 Hz) due to varying loads applied by different Flight Elements
5.7.4	N/A	SMBA-177	Bus voltage requirement became a “should” statement to avoid driving instrument design
7.3.4.6	N/A	SMBA-250	EMI/EMC Testing no longer requires CS116 test due to no identified mechanism for this threat at the satellite level
7.6.1	SMBA-257	SMBA-284	Re-scoped the required TVAC cycles from 12 cycles before launch to 8 cycles before instrument delivery
7.6.2	SMBA-268	SMBA-295	Relaxed TVAC thermal balance requirement from testing of all units to only testing FM1 and any flight units with design changes that affect thermal performance

- Moved to SOW (Deleted from PRD)
 - Section 3.3 Safety and Mission Assurance Requirements
 - Section 5.7.7 Contamination Control Requirements
 - Section 5.7.9 Instrument Simulator and Emulator Requirements
 - Rev C SMBA-119 Instrument Command & Telemetry Electrical GSE
- Restructured sections for clarity and no significant change
 - Section 6.2 EMI/EMC – RF Environment
 - Section 7.3 EMI/EMC Test Requirements*
 - Section 7.4 Structural and Mechanical Test Requirements

*Eliminated CS116 test

- Amendment 18 Industry Day – January 21, 2026
- One on Ones – Requested before January 23, 2026
 - Ongoing until RFP release
- RFP Release – Q1 CY 2026
- Proposals Due – 60 calendar days later
- Award – Q2 CY 2027

Discussion Continues to be Welcome Until Blackout at RFP Release!

Questions?

The background of the slide features a stylized world map. The map is composed of a network of blue dots connected by thin lines, creating a mesh-like structure. The dots are more densely packed in some areas, particularly over the continents, and less dense in others, like the oceans. The overall color scheme is a gradient of dark blue, with the map elements appearing in a slightly lighter shade of blue. The word "Questions?" is written in a large, white, sans-serif font on the left side of the image.